

Ashley Dhar

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EDUCATION

Bachelor of Science in Biomedical Engineering

Expected May 2027

Boston University · GPA: 3.5/4.0 · Dean's List

Boston, MA

Coursework: Organic Chemistry, Molecular & Cell Biology, Circuits, Biomedical Measurements, Signals & Controls

Biomedical Engineering Coursework

Aug 2023 – May 2024

University of Connecticut

Storrs, CT

RESEARCH EXPERIENCE

Undergraduate Research Assistant

Apr 2025 – Present

Dental Biomaterials Laboratory, Housman Medical Research Center, Boston University

Boston, MA

- Characterized chemical composition and functional groups of biomaterial scaffolds via FTIR spectroscopy, establishing structure-property relationships for therapeutic delivery applications
- Performed SEM imaging to analyze nanostructure morphology, porosity, and surface topography, correlating microstructural features with cellular adhesion and material performance
- Conducted mammalian cell culture experiments including MTT viability assays and cytotoxicity screening under aseptic conditions to evaluate biocompatibility of candidate materials
- Investigated bacterial adhesion on dental and denture material surfaces by culturing E. coli in a milk-based medium under mechanical agitation; assessed how surface hardness and corrosion relate to microbial colonization

Student Researcher

Sept 2022 – May 2023

Danbury Hospital, Nuvance Health

Danbury, CT

- Rotated across microbiology, hematology, clinical chemistry, blood bank, diagnostics, and histology divisions; observed workflows and instrumentation including centrifuges, flow cytometers, and automated analyzers in a clinical laboratory setting
- Assisted in processing histopathology specimens using tissue fixation, microtome sectioning, and H&E staining; observed diagnostic workflows alongside pathologists and developed foundational understanding of disease pathology

PROJECTS

Dynamic Light Scattering Analysis of Yogurt Nanoparticles

Spring 2026

Biomedical Measurements (BE493), Boston University

Boston, MA

- Characterized nanoparticle size distributions and polydispersity index across regular, Greek, and drinkable yogurt using a Litesizer; designed sample preparation protocol and dilution series, and evaluated statistical differences across groups using one-way ANOVA with Tukey HSD post-hoc testing

Point-to-Point IR Optical Communication System

Spring 2026

BU Photonics Center

Boston, MA

- Built a wireless IR communication system using Arduino microcontrollers, a 950nm IR LED, and a phototransistor receiver with op-amp signal amplification; implemented pulse-width bit encoding and decoding, troubleshoot signal instability, and improved decoding accuracy from ~30% to ~100% by increasing LED drive current and encoding time

PROFESSIONAL EXPERIENCE

AI Data Annotator

Jan 2025 – Present

RWS Group & Mercor

Remote

- Annotate large datasets for AI/ML model training with high accuracy; developed expertise in data quality assurance and structured labeling methodologies

Pharmacy Associate

Dec 2023 – Sept 2024

CVS Pharmacy

Fairfield, CT

- Assisted pharmacists with prescription processing and general inventory management under DEA compliance and QA protocols; supported patient-facing interactions including medication pickup and basic usage questions

LEADERSHIP & ACTIVITIES

Member, Biomedical Engineering Society (BMES)

Sept 2024 – Present

Boston University

Boston, MA

- Attend industry networking events and panels connecting students with professionals across biomedical research, medical devices, and healthcare; engage with peers across BME concentrations on emerging topics in the field

TECHNICAL SKILLS

Laboratory: FTIR Spectroscopy, SEM, Dynamic Light Scattering (DLS), Mammalian Cell Culture, MTT Assays, Cytotoxicity Testing, Biomaterials Characterization, PCR, Microtomy, Aseptic Technique, Histological Preparation

Computational: MATLAB, Python, Arduino (C), SolidWorks, LaTeX, Molecular Modeling

Languages: English (Native), French (Conversational)

AWARDS

Exemplary Work in Science Research or Engineering, Connecticut STEM Foundation (2023) · Bausch & Lomb Honorary Science Award (2022)